

ELITE IGCSE ACADEMY

Student Past Paper Solutions

May 2023 P1HR Worked Solutions

May 2023 | P1HR

31 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1** **Marks: 3****TOPIC: Prime Factors, HCF & LCM**

May 2023 P1HR - May 2023 | P1HR

- 1** Write 2250 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 1

Marks: 3

TOPIC: Prime Factors, HCF & LCM

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Find the LCM

$$2250 = 225 \times 10$$

$$225 = 3^2 \times 5^2, \quad 10 = 2 \times 5$$

2. Find the LCM

So

$$2250 = 3^2 \times 5^2 \times 2 \times 5$$

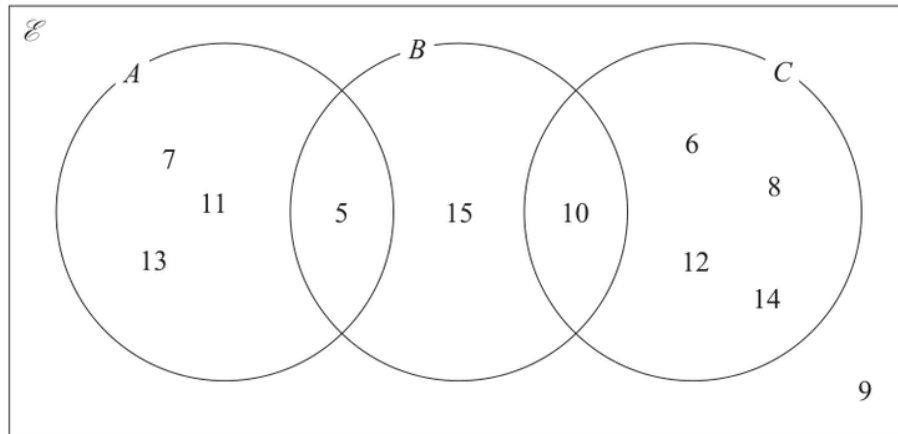
$$2250 = 2 \times 3^2 \times 5^3$$

FINAL ANSWER

$$2 \times 3^2 \times 5^3$$

PAST PAPER QUESTION**Q#: 2****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

May 2023 P1HR - May 2023 | P1HR

2 Here is a Venn diagram.

(a) Write down the numbers that are in the set

(i) A
(1)(ii) $B \cup C$
(1)Dominic writes down $9 \notin C$

(b) Explain why Dominic is correct.

.....
.....
(1)**(Total for Question 2 is 3 marks)**

PAST PAPER SOLUTION

Q#: 2

Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The numbers inside are the

The numbers inside A are the numbers in the A -only region and the overlap with B :

$$A = \{5, 7, 11, 13\}$$

2. Calculate statistic

$B \cup C$ means every number inside B or C :

$$B \cup C = \{5, 6, 8, 10, 12, 14, 15\}$$

3. Dominic writes . This is

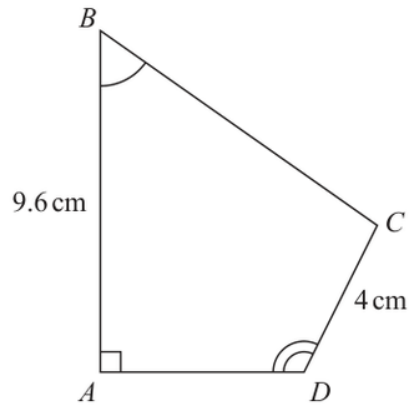
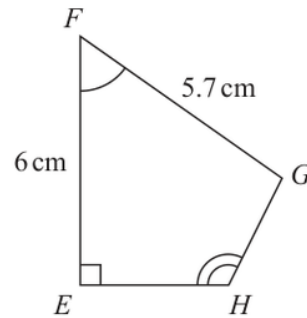
Dominic writes $9 \notin C$. This is correct because 9 is outside circle C in the Venn diagram.

FINAL ANSWER

(a)(i) $\{5, 7, 11, 13\}$, (a)(ii) $\{5, 6, 8, 10, 12, 14, 15\}$. Dominic is correct because 9 is not in C .

PAST PAPER QUESTION**Q#: 3****Marks: 4****TOPIC:** Congruence, Similarity & Geometrical Proof

May 2023 P1HR - May 2023 | P1HR

3 $ABCD$ and $EFGH$ are similar quadrilaterals.Diagram **NOT**
accurately drawn(a) Work out the length of GH cm
(2)(b) Work out the length of BC cm
(2)**(Total for Question 3 is 4 marks)**

PAST PAPER SOLUTION**Q#: 3****Marks: 4****TOPIC:** Congruence, Similarity & Geometrical Proof

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**The scale factor from $EFGH$ to $ABCD$ is

$$\frac{9.6}{6} = 1.6$$

2. Evaluate fractionSince GH corresponds to CD ,

$$GH = \frac{4}{1.6} = 2.5$$

3. Calculate valueSince FG corresponds to BC ,

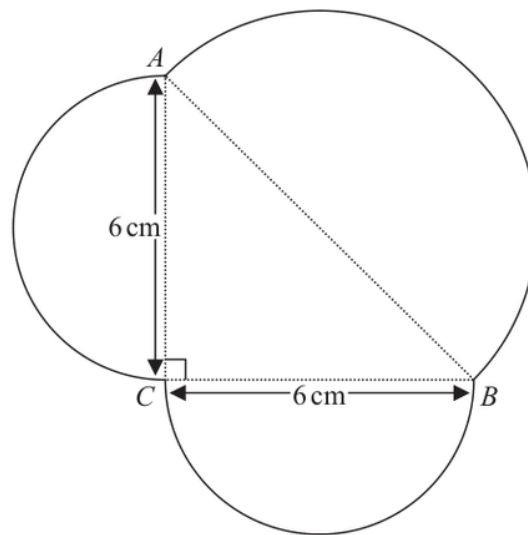
$$BC = 5.7 \times 1.6 = 9.12$$

FINAL ANSWER $GH = 2.5$ cm, and $BC = 9.12$ cm.

PAST PAPER QUESTION**Q#: 4****Marks: 5****TOPIC: Circles, Arcs & Sectors**

May 2023 P1HR - May 2023 | P1HR

- 4 The diagram shows a shape made up of three semicircles, enclosing a right-angled triangle.

Diagram **NOT**
accurately drawn

AB , BC and CA are each the diameter of a semicircle.

$BC = CA = 6$ cm.

Work out the perimeter of the shape.

Give your answer correct to one decimal place.

..... cm

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 5

TOPIC: **Circles, Arcs & Sectors**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Simplify surd

$$AB = \sqrt{6^2 + 6^2} = \sqrt{72} = 8.485 \dots$$

2. The perimeter is the sum

The perimeter is the sum of three semicircular arcs:

$$\frac{\pi}{2}(AB + BC + CA) = \frac{\pi}{2}(8.485 \dots + 6 + 6) = 32.178 \dots$$

FINAL ANSWER

32.2 cm.

EA

PAST PAPER QUESTION**Q#: 5****Marks: 2****TOPIC: Probability Toolkit**

May 2023 P1HR - May 2023 | P1HR

- 5** Each time Evie plays a game against her computer, she will win or lose.

For each game, the probability that Evie will win is 0.74
Evie is going to play 300 games against her computer.

Work out an estimate for the number of games that Evie will lose.

(Total for Question 5 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 5 Marks: 2

TOPIC: **Probability Toolkit**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate probability

$$P(\text{lose}) = 1 - 0.74 = 0.26$$

2. Expected losses

Expected losses:

$$300 \times 0.26 = 78$$

FINAL ANSWER

78.

EA

PAST PAPER QUESTION**Q#: 6****Marks: 4****TOPIC: Algebraic Roots & Indices**

May 2023 P1HR - May 2023 | P1HR

6 (a) Simplify $m^{10} \div m^3$
(1)

$$k^n \times k^4 = k^{12}$$

(b) Write down the value of n $n =$
(1)(c) Simplify $(3x^6y^8)^2$
(2)**(Total for Question 6 is 4 marks)**

PAST PAPER SOLUTION**Q#: 6****Marks: 4****TOPIC: Algebraic Roots & Indices**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

$$(a) m^{10} \div m^3 = m^{10-3} = m^7.$$

2. Use index laws

$$(b) k^n \times k^4 = k^{n+4} = k^{12}, \text{ so } n + 4 = 12 \text{ and } n = 8.$$

3. (c)

$$(c) (3x^6y^8)^2 = 3^2x^{12}y^{16} = 9x^{12}y^{16}.$$

FINAL ANSWER

$$(a) m^7, (b) 8, (c) 9x^{12}y^{16}$$

EA

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Factorising**

May 2023 P1HR - May 2023 | P1HR

7 (a) Expand $4x(x - 5)$
(1)(b) Factorise $y^2 - 9y + 20$
(2)

(Total for Question 7 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 7

Marks: 3

TOPIC: **Factorising**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. (a)

(a)

$$4x(x - 5) = 4x^2 - 20x$$

2. (b)

(b)

$$y^2 - 9y + 20 = (y - 4)(y - 5)$$

FINAL ANSWER

(a) $4x^2 - 20x$, (b) $(y - 4)(y - 5)$

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Powers, Roots & Standard Form**

May 2023 P1HR - May 2023 | P1HR

- 8 (a) Write 5.6×10^{-3} as an ordinary number.

.....
(1)

- (b) Work out $\frac{6 \times 10^3}{2.1 \times 10^{-4} + 9 \times 10^{-5}}$

Give your answer in standard form.

.....
(2)

(Total for Question 8 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Powers, Roots & Standard Form**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert standard form**

$$5.6 \times 10^{-3} = 0.0056$$

2. Convert standard form

For part (b), simplify the denominator:

$$2.1 \times 10^{-4} + 9 \times 10^{-5} = 0.00021 + 0.00009$$

$$= 0.00030 = 3 \times 10^{-4}$$

3. Convert standard form

Then

$$\frac{6 \times 10^3}{3 \times 10^{-4}} = 2 \times 10^{3-(-4)}$$

$$= 2 \times 10^7$$

FINAL ANSWER(a) 0.0056, (b) 2×10^7

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Compound Interest & Depreciation**

May 2023 P1HR - May 2023 | P1HR

- 9 Kazi buys a car for 700 000 taka.
The value of the car depreciates by 12% each year.
- Work out the value of the car at the end of 3 years.
Give your answer correct to the nearest taka.

..... taka

(Total for Question 9 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 9****Marks: 3****TOPIC: Compound Interest & Depreciation**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. A depreciation of gives multiplier**

A depreciation of 12% gives multiplier

$$0.88$$

2. Use compound interest

After 3 years,

$$700000(0.88)^3 = 477030.4$$

3. Use compound interest

Correct to the nearest taka,

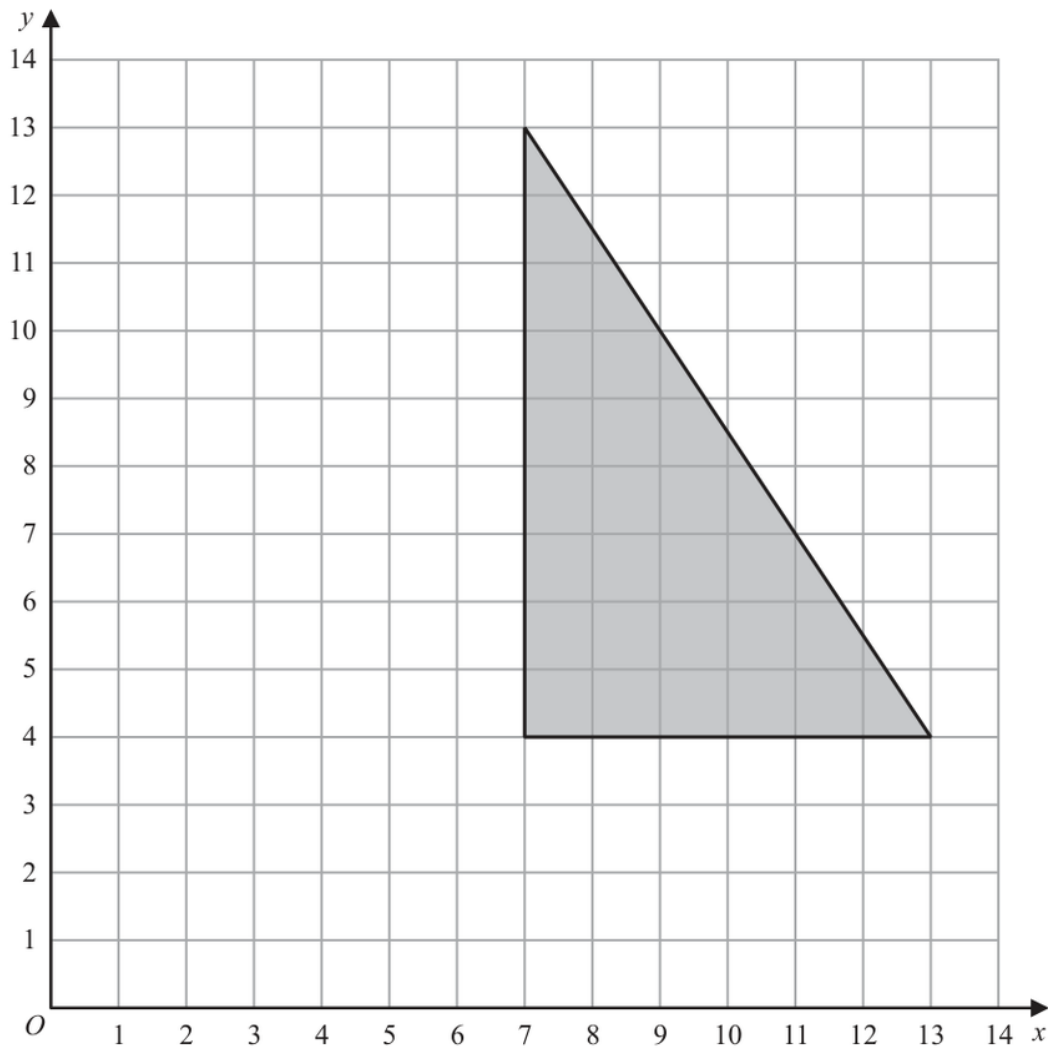
$$477030.4 \approx 477030$$

FINAL ANSWER

477030 taka

PAST PAPER QUESTION**Q#: 10****Marks: 2****TOPIC: Transformations**

May 2023 P1HR - May 2023 | P1HR

10

On the grid, enlarge the shaded shape with scale factor $\frac{1}{3}$ and centre (1, 7)

(Total for Question 10 is 2 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 2****TOPIC: Transformations**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use**

Use

$$\text{new point} = (1, 7) + \frac{1}{3}(\text{old point} - (1, 7))$$

2. The original vertices are

The original vertices are

$$(7, 4), (13, 4), (7, 13)$$

3. They become

They become

$$(3, 6), (5, 6), (3, 9)$$

FINAL ANSWERdraw the triangle with vertices $(3, 6), (5, 6), (3, 9)$.

PAST PAPER QUESTION**Q#: 11****Marks: 3****TOPIC: Standard & Compound Units**

May 2023 P1HR - May 2023 | P1HR

- 11 The diagram shows a block of iron in the shape of a cuboid.

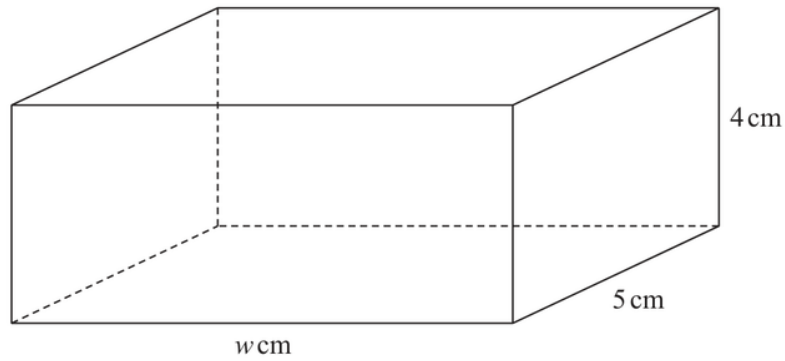


Diagram **NOT**
accurately drawn

The block has length w cm, width 5 cm and height 4 cm

The density of iron is 7.8 g/cm^3

The mass of the block is 1950 g

Work out the value of w

$w = \dots\dots\dots$

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 3****TOPIC: Standard & Compound Units**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate volume**

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$7.8 = \frac{1950}{\text{volume}}$$

$$\text{volume} = \frac{1950}{7.8} = 250 \text{ cm}^3$$

2. Calculate value

For the cuboid,

$$w \times 5 \times 4 = 250$$

$$20w = 250$$

$$w = 12.5$$

FINAL ANSWER

$$w = 12.5 \text{ cm.}$$

PAST PAPER QUESTION**Q#: 12****Marks: 5****TOPIC: Combined & Conditional Probability**

May 2023 P1HR - May 2023 | P1HR

12 Moeen has a biased 6-sided dice.

The table gives information about the probability that, when the dice is thrown, it will land on each number.

Number	1	2	3	4	5	6
Probability	x	0.15	0.5	y	0.13	0.03

(a) Show that $x + y = 0.19$

(2)

Given that $3x - y = 0.09$ and $x + y = 0.19$ (b) work out the value of x and the value of y
Show clear algebraic working. $x = \dots\dots\dots$ $y = \dots\dots\dots$

(3)

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 5****TOPIC: Combined & Conditional Probability**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate probability**

The probabilities add to 1:

$$x + 0.15 + 0.50 + y + 0.13 + 0.03 = 1$$

$$x + y = 0.19$$

2. Given

Given

$$3x - y = 0.09$$

3. Add the two equations

Add the two equations:

$$4x = 0.28$$

$$x = 0.07$$

$$y = 0.19 - 0.07 = 0.12$$

FINAL ANSWER

$$x = 0.07, y = 0.12.$$

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

May 2023 P1HR - May 2023 | P1HR

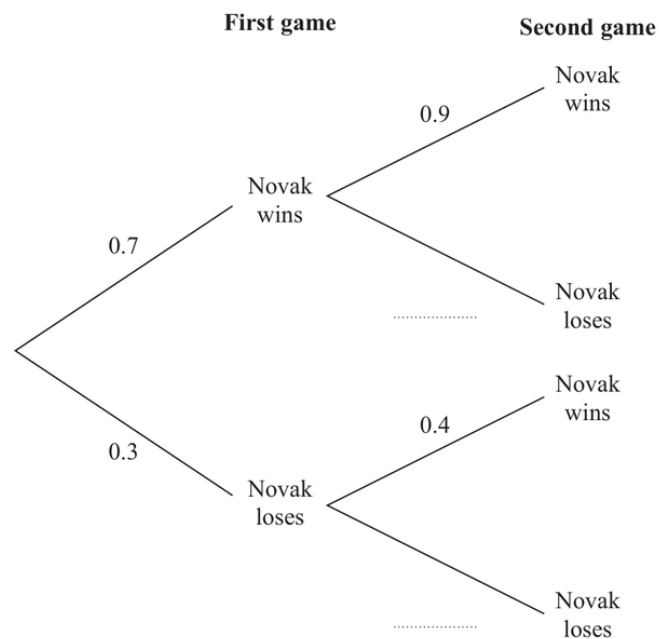
13 Novak is going to play two games of tennis.

The probability that he wins the first game is 0.7

If he wins the first game, the probability that he wins the second game is 0.9

If he loses the first game, the probability that he wins the second game is 0.4

(a) Complete the probability tree diagram.



(1)

(b) Work out the probability that Novak wins both games of tennis.

(2)

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13

Marks: 3

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The missing branch probabilities are

The missing branch probabilities are

$$0.1, \quad 0.6$$

2. Calculate probability

Novak wins both games with probability

$$0.7 \times 0.9 = 0.63$$

FINAL ANSWER

0.63.

EA

PAST PAPER QUESTION**Q#: 14****Marks: 6****TOPIC: Cumulative Frequency Diagrams**

May 2023 P1HR - May 2023 | P1HR

14 The table gives information about the times taken by 80 people to run a race.

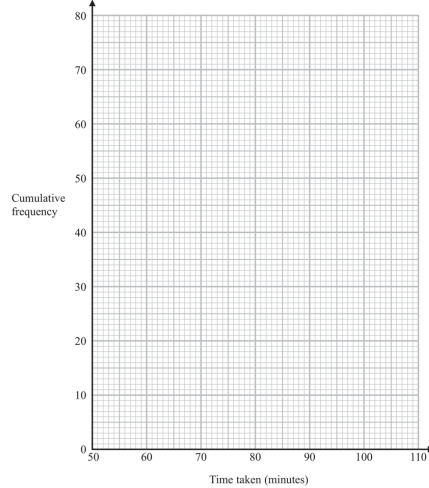
Time taken (t minutes)	Frequency
$50 < t \leq 60$	15
$60 < t \leq 70$	16
$70 < t \leq 80$	21
$80 < t \leq 90$	14
$90 < t \leq 100$	8
$100 < t \leq 110$	6

(a) Complete the cumulative frequency table.

Time taken (t minutes)	Cumulative frequency
$50 < t \leq 60$	
$60 < t \leq 70$	
$70 < t \leq 80$	
$80 < t \leq 90$	
$90 < t \leq 100$	
$100 < t \leq 110$	

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the median time taken.

_____ minutes
(1)

(d) Use your graph to find an estimate for the interquartile range of the times taken.

_____ minutes
(2)

(Total for Question 14 is 6 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 6****TOPIC: Cumulative Frequency Diagrams**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The cumulative frequencies are**

The cumulative frequencies are

$$15, 31, 52, 66, 74, 80$$

2. Convert standard form

For 80 runners,

$$\text{median} = 40, \quad Q_1 = 20, \quad Q_3 = 60$$

$$\text{median} = 70 + \frac{40 - 31}{52 - 31} \times 10 \approx 74.3$$

$$Q_1 = 60 + \frac{20 - 15}{31 - 15} \times 10 \approx 63.1$$

$$Q_3 = 80 + \frac{60 - 52}{66 - 52} \times 10 \approx 85.7$$

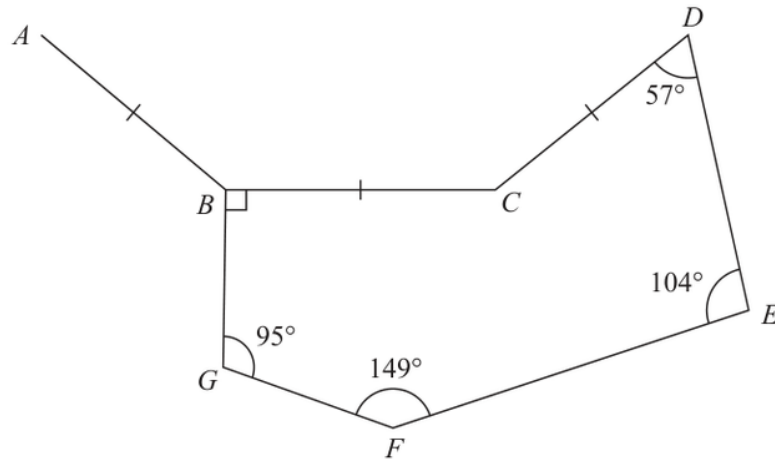
$$\text{IQR} \approx 85.7 - 63.1 = 22.6$$

FINAL ANSWER

median about 74.3 min, IQR about 22.6 min.

PAST PAPER QUESTION**Q#: 15****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

May 2023 P1HR - May 2023 | P1HR

15Diagram **NOT**
accurately drawn $BCDEFG$ is a hexagon. AB , BC and CD are three sides of a regular n -sided polygon.Calculate the value of n
Show your working clearly. $n = \dots\dots\dots$ **(Total for Question 15 is 4 marks)**

PAST PAPER SOLUTION**Q#: 15****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The sum of the interior**

The sum of the interior angles of a hexagon is

$$(6 - 2) \times 180 = 720^\circ$$

2. Find the gradientThe reflex angle at C in the hexagon is

$$720^\circ - (90^\circ + 95^\circ + 149^\circ + 104^\circ + 57^\circ) = 225^\circ$$

3. Find the gradientSo the smaller angle BCD is

$$360^\circ - 225^\circ = 135^\circ$$

4. This is the interior angleThis is the interior angle of the regular n -sided polygon.**5. Find the gradient**

So the exterior angle is

$$180^\circ - 135^\circ = 45^\circ$$

$$n = \frac{360}{45} = 8$$

FINAL ANSWER

$$n = 8.$$

PAST PAPER QUESTION**Q#: 16****Marks: 2**TOPIC: **Algebraic Proof**

May 2023 P1HR - May 2023 | P1HR

16 Use algebra to show that $0.1\dot{7}\dot{6} = \frac{35}{198}$

(Total for Question 16 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 16****Marks: 2****TOPIC: Algebraic Proof**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let**

Let

$$x = 0.1767676 \dots$$

2. Calculate value

Then

$$10x = 1.767676 \dots$$

3. Calculate value

and

$$1000x = 176.767676 \dots$$

4. Subtract

Subtract:

$$990x = 175$$

$$x = \frac{175}{990} = \frac{35}{198}$$

FINAL ANSWER

$$0.1767676 \dots = \frac{35}{198}$$

PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Direct & Inverse Proportion**

May 2023 P1HR - May 2023 | P1HR

17 F is inversely proportional to the square of r

$$F = 36 \text{ when } r = 4$$

(a) Find a formula for F in terms of r

.....
(3)

(b) Work out the value of F when $r = 48$

.....
(1)

(Total for Question 17 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 17****Marks: 4****TOPIC: Direct & Inverse Proportion**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find inverse function** F is inversely proportional to r^2 , so

$$F = \frac{k}{r^2}$$

2. Use whenUse $F = 36$ when $r = 4$:

$$36 = \frac{k}{4^2}$$

$$k = 36 \times 16 = 576$$

3. Find inverse function

So

$$F = \frac{576}{r^2}$$

4. Find inverse functionWhen $r = 48$,

$$F = \frac{576}{48^2} = \frac{576}{2304} = 0.25$$

FINAL ANSWER $F = \frac{576}{r^2}$, and $F = 0.25$ when $r = 48$.

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

May 2023 P1HR - May 2023 | P1HR

18 Diego builds a fence using fence panels.

The total length of the fence is 50 metres, correct to the nearest 5 metres.

The length of each fence panel is 1.3 metres, correct to the nearest 10 cm.

The cost of each fence panel is £8.65

Diego may only buy complete fence panels.

Diego only pays for the number of panels he needs to build the fence.

Work out the greatest difference in the possible amounts that Diego could pay to build the fence.

Show your working clearly.

£.....

(Total for Question 18 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 18****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The fence length m correct**

The fence length 50 m correct to the nearest 5 m has bounds

$$47.5 \leq L < 52.5$$

2. The panel length m correct

The panel length 1.3 m correct to the nearest 10 cm has bounds

$$1.25 \leq p < 1.35$$

3. Smallest possible number of panels

Smallest possible number of panels:

$$\left\lceil \frac{47.5}{1.35} \right\rceil = 36$$

4. Greatest possible number of panels

Greatest possible number of panels:

$$\left\lfloor \frac{52.5}{1.25} \right\rfloor = 42$$

5. Difference in number of panels

Difference in number of panels:

$$42 - 36 = 6$$

6. Greatest difference in cost

Greatest difference in cost:

$$6 \times 8.65 = 51.90$$

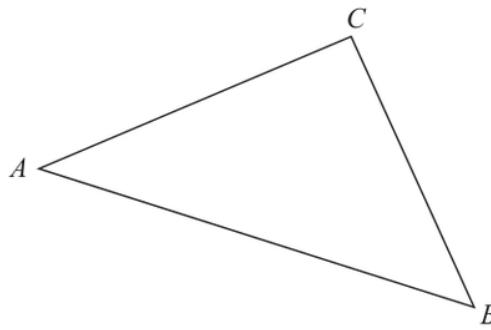
FINAL ANSWER

51.90 pounds

PAST PAPER QUESTION**Q#: 19****Marks: 5**TOPIC: **Vectors**

May 2023 P1HR - May 2023 | P1HR

- 19** The diagram shows a triangle ABC where A , B and C represent the positions of three towns.

Diagram **NOT**
accurately drawn

$$\vec{AB} = \begin{pmatrix} 7 \\ -2 \end{pmatrix} \quad \vec{BC} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$$

Pru travels directly from A to B and then directly from B to C

Yang travels directly from A to C

Given that the values for $\vec{AB}$ and $\vec{BC}$ are in kilometres,

work out how much further Pru travels than Yang travels.

Give your answer in km, correct to one decimal place.

..... km

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 5**TOPIC: **Vectors**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Pru travels**

Pru travels

$$|AB| + |BC| = \sqrt{7^2 + (-2)^2} + \sqrt{(-3)^2 + 5^2}$$

$$= \sqrt{53} + \sqrt{34} = 13.111\dots$$

2. Simplify surdYang travels from A to C .

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \begin{pmatrix} 7 \\ -2 \end{pmatrix} + \begin{pmatrix} -3 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$|AC| = \sqrt{4^2 + 3^2} = 5$$

3. Difference

Difference:

$$13.111\dots - 5 = 8.111\dots$$

FINAL ANSWER

Pru travels 8.1 km further.

PAST PAPER QUESTION**Q#: 20****Marks: 7****TOPIC: Transformations of Graphs**

May 2023 P1HR - May 2023 | P1HR

20 Curve **C** has equation $y = f(x)$ The graph of curve **C** has one maximum point.

The coordinates of this maximum point are (3, 5)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = 2f(x)$

(.....,)
(1)

(ii) $y = f(x) - 7$

(.....,)
(1)

(iii) $y = f(-x)$

(.....,)
(1)Curve **L** has equation $y = x^2 + 7x + 20$ Curve **L** is transformed to curve **S** under the translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ (b) Find an equation for **S**Give your answer in the form $y = ax^2 + bx + c$ $y = \dots\dots\dots$
(4)**(Total for Question 20 is 7 marks)**

PAST PAPER SOLUTION**Q#: 20****Marks: 7****TOPIC: Transformations of Graphs**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The maximum point of is**The maximum point of $y = f(x)$ is $(3, 5)$.**2. Calculate value**

For

$$y = 2f(x)$$

3. the coordinate is multiplied bythe y -coordinate is multiplied by 2:

$$(3, 5) \rightarrow (3, 10)$$

4. Calculate value

For

$$y = f(x) - 7$$

5. the graph moves 7 units

the graph moves 7 units down:

$$(3, 5) \rightarrow (3, -2)$$

6. Calculate value

For

$$y = f(-x)$$

7. the graph reflects in thethe graph reflects in the y -axis:

$$(3, 5) \rightarrow (-3, 5)$$

8. Calculate value

For part (b),

$$y = x^2 + 7x + 20$$

9. The translation vector is

The translation vector is

$$\begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

10. Read the graph

So the graph moves 2 units right.

11. Replace by

Replace x by $x - 2$:

$$y = (x - 2)^2 + 7(x - 2) + 20$$

$$y = x^2 + 3x + 10$$

FINAL ANSWER

$(3, 10)$, $(3, -2)$, $(-3, 5)$, and $y = x^2 + 3x + 10$

EA

PAST PAPER QUESTION**Q#: 20****Marks: 7****TOPIC: Transformations of Graphs**

May 2023 P1HR - May 2023 | P1HR

20 Curve **C** has equation $y = f(x)$ The graph of curve **C** has one maximum point.

The coordinates of this maximum point are (3, 5)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = 2f(x)$

(.....,)
(1)

(ii) $y = f(x) - 7$

(.....,)
(1)

(iii) $y = f(-x)$

(.....,)
(1)Curve **L** has equation $y = x^2 + 7x + 20$ Curve **L** is transformed to curve **S** under the translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ (b) Find an equation for **S**Give your answer in the form $y = ax^2 + bx + c$ $y = \dots\dots\dots$
(4)**(Total for Question 20 is 7 marks)**

PAST PAPER SOLUTION**Q#: 20****Marks: 7****TOPIC: Transformations of Graphs**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The maximum point of is**The maximum point of $y = f(x)$ is $(3, 5)$.**2. Calculate value**

For

$$y = 2f(x)$$

3. the coordinate is multiplied bythe y -coordinate is multiplied by 2:

$$(3, 5) \rightarrow (3, 10)$$

4. Calculate value

For

$$y = f(x) - 7$$

5. the graph moves 7 units

the graph moves 7 units down:

$$(3, 5) \rightarrow (3, -2)$$

6. Calculate value

For

$$y = f(-x)$$

7. the graph reflects in thethe graph reflects in the y -axis:

$$(3, 5) \rightarrow (-3, 5)$$

8. Calculate value

For part (b),

$$y = x^2 + 7x + 20$$

9. The translation vector is

The translation vector is

$$\begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

10. Read the graph

So the graph moves 2 units right.

11. Replace by

Replace x by $x - 2$:

$$y = (x - 2)^2 + 7(x - 2) + 20$$

$$y = x^2 + 3x + 10$$

FINAL ANSWER

$(3, 10)$, $(3, -2)$, $(-3, 5)$, and $y = x^2 + 3x + 10$

EA

PAST PAPER QUESTION**Q#: 21****Marks: 6****TOPIC: Differentiation**

May 2023 P1HR - May 2023 | P1HR

21 The curve **T** has equation $y = x^3 - 2x^2 - 9x + 15$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

(2)

(b) Find the range of values of x for which **T** has a positive gradient.
Give your values correct to 3 significant figures.
Show your working clearly.

(4)

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 6****TOPIC: Differentiation**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$y = x^3 - 2x^2 - 9x + 15$$

$$\frac{dy}{dx} = 3x^2 - 4x - 9$$

2. (b) The curve has a

(b) The curve has a positive gradient when

$$3x^2 - 4x - 9 > 0$$

3. Solve

Solve

$$3x^2 - 4x - 9 = 0$$

$$x = \frac{4 \pm \sqrt{16 + 108}}{6}$$

$$x = \frac{2 \pm \sqrt{31}}{3}$$

4. Differentiate

To 3 significant figures, the roots are

$$x = -1.19, \quad x = 2.52$$

5. Solve quadratic equation

Since the quadratic opens upwards, it is positive outside the roots.

FINAL ANSWER

$$(a) \frac{dy}{dx} = 3x^2 - 4x - 9, \quad (b) \quad x < -1.19 \text{ or } x > 2.52$$

PAST PAPER QUESTION**Q#: 21****Marks: 6****TOPIC: Differentiation**

May 2023 P1HR - May 2023 | P1HR

21 The curve **T** has equation $y = x^3 - 2x^2 - 9x + 15$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

(2)

(b) Find the range of values of x for which **T** has a positive gradient.
Give your values correct to 3 significant figures.
Show your working clearly.

(4)

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 6****TOPIC: Differentiation**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$y = x^3 - 2x^2 - 9x + 15$$

$$\frac{dy}{dx} = 3x^2 - 4x - 9$$

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FINAL ANSWER

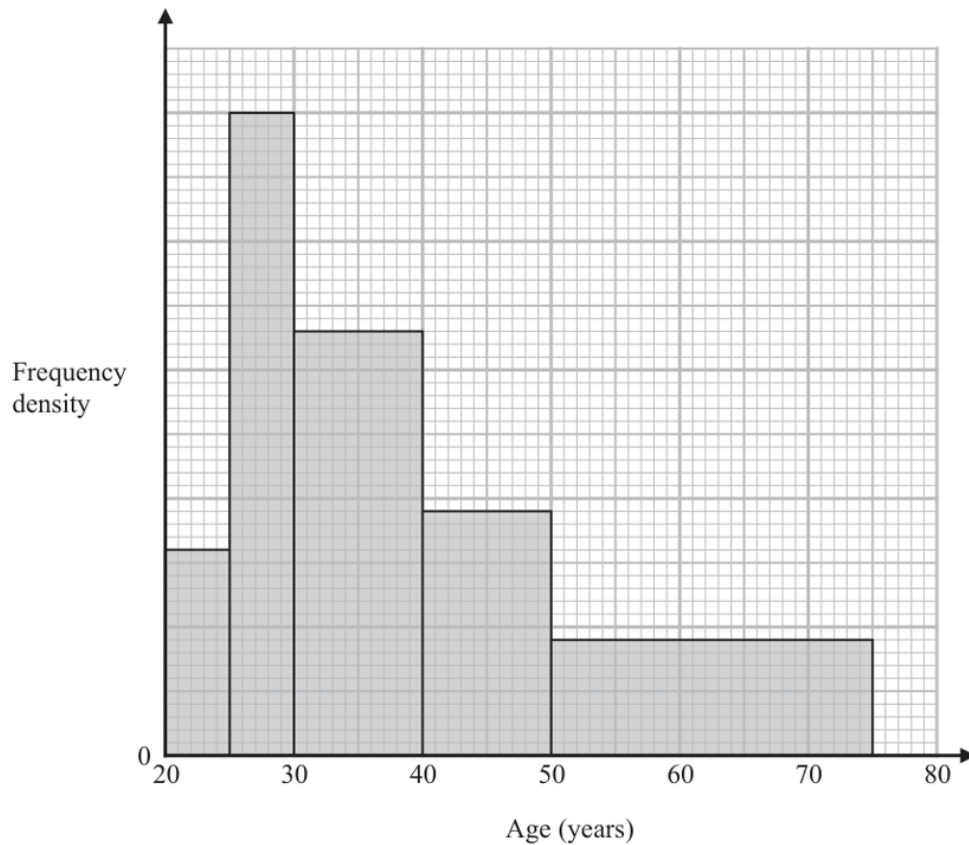
$$(a) \frac{dy}{dx} = 3x^2 - 4x - 9, \quad (b) \quad x < -1.19 \text{ or } x > 2.52$$

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Histograms**

May 2023 P1HR - May 2023 | P1HR

22 Some people attend a concert.

The histogram shows information about the ages of these people.



Work out an estimate for the percentage of these people who are aged more than 55 years.

Give your answer correct to one decimal place.

.....%

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: Histograms**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The scale of the frequency**

The scale of the frequency density axis is not needed because the percentage is found from areas.

2. Calculate area

Using the bar heights from the histogram, the relative areas are:

$$\begin{aligned} & (5)(277) + (5)(866) + (10)(571) + (10)(329) + (25)(155) \\ &= 1385 + 4330 + 5710 + 3290 + 3875 \\ &= 18590 \end{aligned}$$

3. Calculate value

For ages more than 55, use the part of the last bar from 55 to 75:

$$(20)(155) = 3100$$

4. Calculate percentage

So the estimated percentage is

$$\frac{3100}{18590} \times 100 = 16.675 \dots$$

FINAL ANSWER

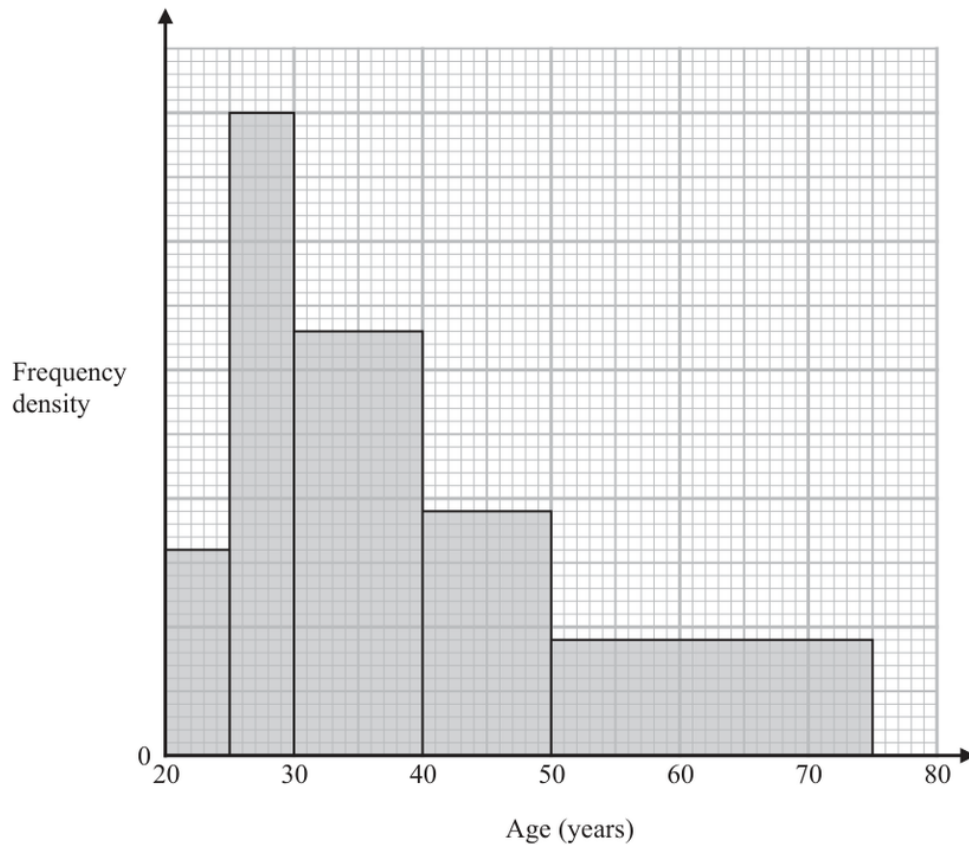
16.7%

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Histograms**

May 2023 P1HR - May 2023 | P1HR

22 Some people attend a concert.

The histogram shows information about the ages of these people.



Work out an estimate for the percentage of these people who are aged more than 55 years.

Give your answer correct to one decimal place.

.....%

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: Histograms**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The scale of the frequency**

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Using the bar heights from the histogram, the relative areas are:

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For ages more than 55, use the part of the last bar from 55 to 75:

$$(20)(155) = 3100$$

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So the estimated percentage is

$$\frac{3100}{18590} \times 100 = 16.675 \dots$$

FINAL ANSWER

16.7%

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Algebraic Fractions**

May 2023 P1HR - May 2023 | P1HR

23 Simplify $(x^2 - 4) \div \left(\frac{4x^2 - 7x - 2}{x} \right) - 2x$

Give your answer in the form $\frac{ax^2}{bx + c}$ where a , b and c are integers.

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Algebraic Fractions**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

$$(x^2 - 4) \div \left(\frac{4x^2 - 7x - 2}{x} \right) - 2x$$

2. Change division into multiplication by

Change division into multiplication by the reciprocal:

$$(x^2 - 4) \times \frac{x}{4x^2 - 7x - 2} - 2x$$

3. Factorise

Factorise:

$$x^2 - 4 = (x - 2)(x + 2)$$

$$4x^2 - 7x - 2 = (4x + 1)(x - 2)$$

4. Simplify surd

So

$$\frac{x(x - 2)(x + 2)}{(4x + 1)(x - 2)} - 2x = \frac{x(x + 2)}{4x + 1} - 2x$$

$$\frac{x(x + 2) - 2x(4x + 1)}{4x + 1} = \frac{x^2 + 2x - 8x^2 - 2x}{4x + 1} = \frac{-7x^2}{4x + 1}$$

FINAL ANSWER

$$\frac{-7x^2}{4x + 1}$$

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Algebraic Fractions**

May 2023 P1HR - May 2023 | P1HR

23 Simplify $(x^2 - 4) \div \left(\frac{4x^2 - 7x - 2}{x} \right) - 2x$

Give your answer in the form $\frac{ax^2}{bx + c}$ where a , b and c are integers.

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Algebraic Fractions**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

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2. Change division into multiplication by

Change division into multiplication by the reciprocal:

$$(x^2 - 4) \times \frac{x}{4x^2 - 7x - 2} - 2x$$

3. Factorise

Factorise:

$$x^2 - 4 = (x - 2)(x + 2)$$

$$4x^2 - 7x - 2 = (4x + 1)(x - 2)$$

4. Simplify surd

So

$$\frac{x(x - 2)(x + 2)}{(4x + 1)(x - 2)} - 2x = \frac{x(x + 2)}{4x + 1} - 2x$$

$$\frac{x(x + 2) - 2x(4x + 1)}{4x + 1} = \frac{x^2 + 2x - 8x^2 - 2x}{4x + 1} = \frac{-7x^2}{4x + 1}$$

FINAL ANSWER

$$\frac{-7x^2}{4x + 1}$$

PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Algebraic Roots & Indices**

May 2023 P1HR - May 2023 | P1HR

24 Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for x in terms of n State any restrictions on n

(Total for Question 24 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 24****Marks: 5****TOPIC: Algebraic Roots & Indices**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use index laws**

$$16^x = 2^{4x}, \quad 8 = 2^3$$

2. Use index laws

So

$$2^n = 2^{x^2} \times 2^{4x} \times 2^3 = 2^{x^2+4x+3}$$

3. Hence

Hence

$$n = x^2 + 4x + 3 = (x + 2)^2 - 1$$

$$(x + 2)^2 = n + 1$$

4. Simplify surdSince $x > 0$,

$$x + 2 = \sqrt{n + 1}$$

$$x = -2 + \sqrt{n + 1}$$

5. Simplify surdFor $x > 0$, $\sqrt{n + 1} > 2$, so $n > 3$.**FINAL ANSWER**

$$x = -2 + \sqrt{n + 1}, \quad n > 3$$

PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Algebraic Roots & Indices**

May 2023 P1HR - May 2023 | P1HR

24 Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

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find an expression for x in terms of n State any restrictions on n

(Total for Question 24 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 24****Marks: 5****TOPIC: Algebraic Roots & Indices**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$2^n = 2^{x^2} \times 16^x \times 8$$

$$16^x = (2^4)^x = 2^{4x}$$

$$8 = 2^3$$

2. Calculate value

So

$$2^n = 2^{x^2+4x+3}$$

$$n = x^2 + 4x + 3$$

3. Complete the square

Complete the square:

$$n = (x + 2)^2 - 1$$

$$(x + 2)^2 = n + 1$$

4. Simplify surdSince $x > 0$, take the positive root:

$$x + 2 = \sqrt{n + 1}$$

$$x = \sqrt{n + 1} - 2$$

5. Simplify surdFor $x > 0$,

$$\sqrt{n + 1} > 2$$

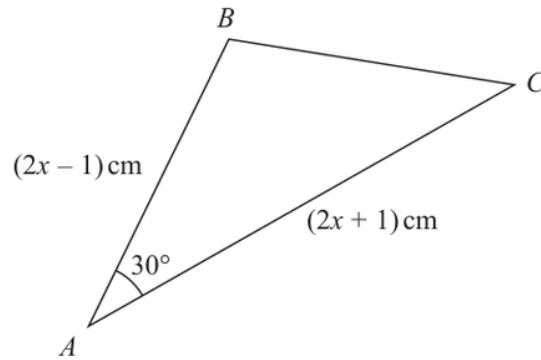
$$n > 3$$

FINAL ANSWER

$$x = \sqrt{n + 1} - 2, \text{ with } n > 3$$

PAST PAPER QUESTION**Q#: 25****Marks: 6****TOPIC: Sine, Cosine Rule & Area of Triangles**

May 2023 P1HR - May 2023 | P1HR

25 Here is a triangle ABC Diagram **NOT**
accurately drawnThe area of the triangle is $(x^2 + x - 3.75) \text{ cm}^2$ Find the size of the largest angle in triangle ABC

Give your answer correct to the nearest degree.

.....
o

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 6****TOPIC: Sine, Cosine Rule & Area of Triangles**

May 2023 P1HR - May 2023 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The two sides around the**The two sides around the 30° angle are

$$2x - 1 \quad \text{and} \quad 2x + 1$$

2. Use trigonometry

Using the area formula,

$$\text{area} = \frac{1}{2}ab \sin C$$

$$x^2 + x - 3.75 = \frac{1}{2}(2x - 1)(2x + 1) \sin 30^\circ$$

3. Use trigonometrySince $\sin 30^\circ = \frac{1}{2}$,

$$x^2 + x - 3.75 = \frac{1}{4}(2x - 1)(2x + 1)$$

$$x^2 + x - 3.75 = \frac{1}{4}(4x^2 - 1)$$

$$x^2 + x - 3.75 = x^2 - 0.25$$

$$x = 3.5$$

4. Use trigonometry

So

$$AB = 2x - 1 = 6, \quad AC = 2x + 1 = 8$$

5. Use cosine ruleUse the cosine rule to find BC :

$$BC^2 = 6^2 + 8^2 - 2(6)(8) \cos 30^\circ$$

$$BC = 4.1063 \dots$$

6. Use trigonometryThe largest side is $AC = 8$, so the largest angle is $\angle B$.**7. Use trigonometry**

Using the cosine rule again:

$$\cos B = \frac{6^2 + BC^2 - 8^2}{2(6)(BC)}$$

$$B = 103.0643\dots^\circ$$

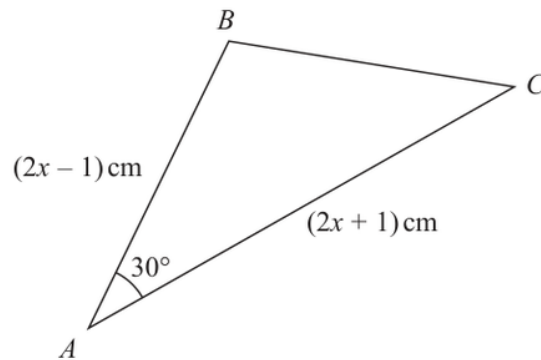
FINAL ANSWER

103°

EA

PAST PAPER QUESTION**Q#: 25****Marks: 6****TOPIC: Sine, Cosine Rule & Area of Triangles**

May 2023 P1HR - May 2023 | P1HR

25 Here is a triangle ABC Diagram **NOT**
accurately drawnThe area of the triangle is $(x^2 + x - 3.75) \text{ cm}^2$ Find the size of the largest angle in triangle ABC
Give your answer correct to the nearest degree.

o

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 6****TOPIC: Sine, Cosine Rule & Area of Triangles**

May 2023 P1HR - May 2023 | P1HR

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FINAL ANSWER

103°

EA